RAHUL R

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About Me

Computer Science graduate specializing in Artificial Intelligence, passionate about transforming innovative ideas into impactful technology solutions. Proficient in Python, machine learning, backend development, and cloud technologies, with hands-on experience in developing intelligent applications across diverse domains. Experienced in research-driven projects, including IEEE-published work. A curious problem-solver driven by continuous learning, creativity, and a strong commitment to building efficient, scalable, and meaningful technological solutions.

Skills

- **Programming:** Python, C, Java, SQL, JavaScript, HTML, CSS
- **Databases and Visualisation:** MySQL, PostgreSQL , MongoDB, Tableau, Power BI, Streamlit
- **AI & ML:** Supervised/Unsupervised Learning, Deep Learning, NLP, Generative AI
- **Libraries and Frameworks:** Pandas, NumPy, Scikit-learn, TensorFlow, Keras
- **Soft Skills:** Communication, Leadership, Quick Learner, Problem-Solving

Projects

- **AI-Enabled Patient Health Monitoring System(IEEE Published)**
  Python, Random Forest, SHAP, AES-256-EAX, Generative AI, Streamlit
  - **Predictive Analytics:** Developed dual Random Forest models for patient vital anomaly detection and diabetes risk prediction.
  - **Security & Explainability:** Implemented AES-256-EAX encryption and SHAP-based visualizations for secure and interpretable predictions.
  - **Generative AI:** Integrated an LLM assistant to analyze patient vitals, EHR, and medication records to generate personalized emergency protocols.
  - **Dashboard & Performance:** Built a Streamlit clinical dashboard with patient monitoring and analytics, achieving 98% accuracy in vital anomaly detection.
- **Calories Burnt Prediction Using XGBoost Regression**
  Python, Machine Learning, XGBoost, Regression Analysis
  - **Predictive Modeling:** Developed a machine learning model using XGBoost Regression to predict calories burnt based on user activity and physiological data.
  - **Model Optimization:** Applied gradient boosting techniques to improve prediction performance and reduce model errors by 18%.
  - **Feature Engineering:** Analyzed and extracted relevant features from activity datasets to improve prediction reliability by 15%.
  - **Performance Evaluation:** Achieved 92% prediction accuracy, demonstrating the effectiveness of ensemble learning for fitness analytics.
- **IoT Solar Power Monitoring and Management System**
  ESP32, IoT, Cloud Storage, Monitoring Systems
  - **IoT-Based Monitoring:** Developed an ESP32-based solar power monitoring solution to track system performance and improve monitoring accuracy by 25%.
  - **Performance Optimization:** Implemented monitoring and fault detection mechanisms to identify system irregularities and minimize downtime by 40%.
  - **Cloud Integration:** Integrated cloud storage solutions for efficient data management, monitoring, and accessibility.
  - **System Reliability:** Contributed to achieving 99.9% system uptime through performance optimization and efficient resource management.
- **AI-Powered Driver Drowsiness Detection System**
  Python, OpenCV, Computer Vision, Deep Learning
  - **Real-Time Face Detection:** Implemented real-time facial detection using computer vision techniques to identify and track driver faces.
  - **Facial Landmark Analysis:** Monitored facial landmarks and eye-related features to identify visual indicators of driver fatigue and drowsiness.
  - **Deep Learning Integration:** Integrated a pre-trained deep learning model to classify drowsiness conditions and support automated detection.
  - **Driver Safety:** Developed a monitoring approach aimed at identifying potential drowsiness events and improving road safety through early detection.

Experience

- **Next24tech Technology & Services** 2024
  Web Development Intern
  - **Web Development:** Developed responsive web pages using HTML, CSS, JavaScript, and React.
  - **Team Collaboration:** Collaborated with the development team to improve website functionality and user experience.
  - **Technical Experience:** Gained hands-on experience in front-end and full-stack web development.

Education

- **Karunya Institute of Technology and Sciences** Coimbatore, India
  Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence) *2022 – 2026*
  - CGPA: 8.94
- **Bhavans Vidya Mandir** Palakkad, India
  Higher Secondary *2021*
  - Percentage: 91%

Certifications

- IoT Fundamentals: Connected Things
- PCAP: Programming Essentials in Python
- Java Programming Fundamentals – Infosys
- Introduction to Artificial Intelligence – Simplilearn
- CLA – Programming in Problem Solving
- Google Cybersecurity – Coursera
- Network Addressing and Basic Troubleshooting – Cisco

Languages

- English
- Hindi
- Malayalam(Native Speaker)